EMERALD PRELOVED MARKETPLACE MOBILE APPLICATION

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1. Feedback Implementation and Revisions

In order to align this report with the Excellent criteria specified in the Assignment 2 rubric, a number of targeted changes were made after the Assignment 1 evaluation, which resulted in a mark of 66% .
First, the lecturer emphasized that earlier answers were too unclear to meet academic criteria and lacked adequate depth and critical engagement. This study uses an organized "Why-How-Who" analytical approach in response. According to Rogers, Sharp, and Preece (2023), p. 495, this guarantees that each design choice is not only explained but also firmly grounded in established interaction design theory, elucidating the psychological motivation behind the UI, the technical implementation details, and the particular user group that benefits.

Second, the first submission's ambiguous screenshots have been fixed. Each design approach is clearly connected to labelled low-fidelity prototype figures in the current study. This guarantees that the application of theory is observable and verifiable by the marker, eliminating any possibility of speculation about the status of the system in other words they are now clearly visible.

Lastly, this report substitutes five completely new design principles from Learning Unit 6: Affordance, Mapping, Simplicity, Findability, and Navigability for the principles used in Assignment 1: Constraints, Consistency, Visibility, and Feedback. This time, I have made sure to be thorough in my explanations and to include the reasons and methods for putting these design principles into practice. According to the standards of the 2025–2026 academic cycle, this decision shows a more comprehensive understanding of the interaction design sector (Rogers, Sharp, and Preece, 2023, p. 27).

2. Design Principle 1: Affordance

Definition and Motivation (Why): According to Rogers, Sharp, and Preece (2023), p. 31, affordance refers to an object's characteristics that show how it might be utilized; it essentially gives a clue to the user. Perceived affordances, which are simply learnt norms where graphical features indicate an action, are used by designers in digital systems (Rogers, Sharp, and Preece, 2023, p. 31).

Implementation (How): Elevated rectangular card components (A) are used to represent textbook listings, as seen in Figure 1 (Low-Fidelity Home Dashboard Prototype). To set these cards apart from the background, they use well-defined edges and faint shadows. The interface gives a visual indication that the element is a distinct, interactive object by imitating the physical shape of a card or book.

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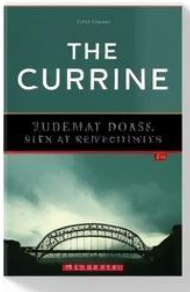

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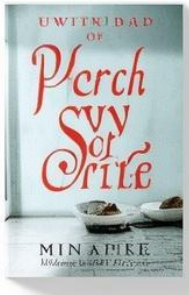

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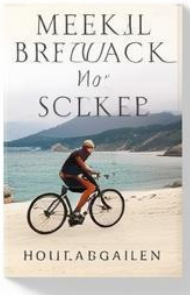

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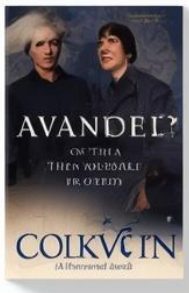

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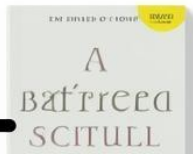
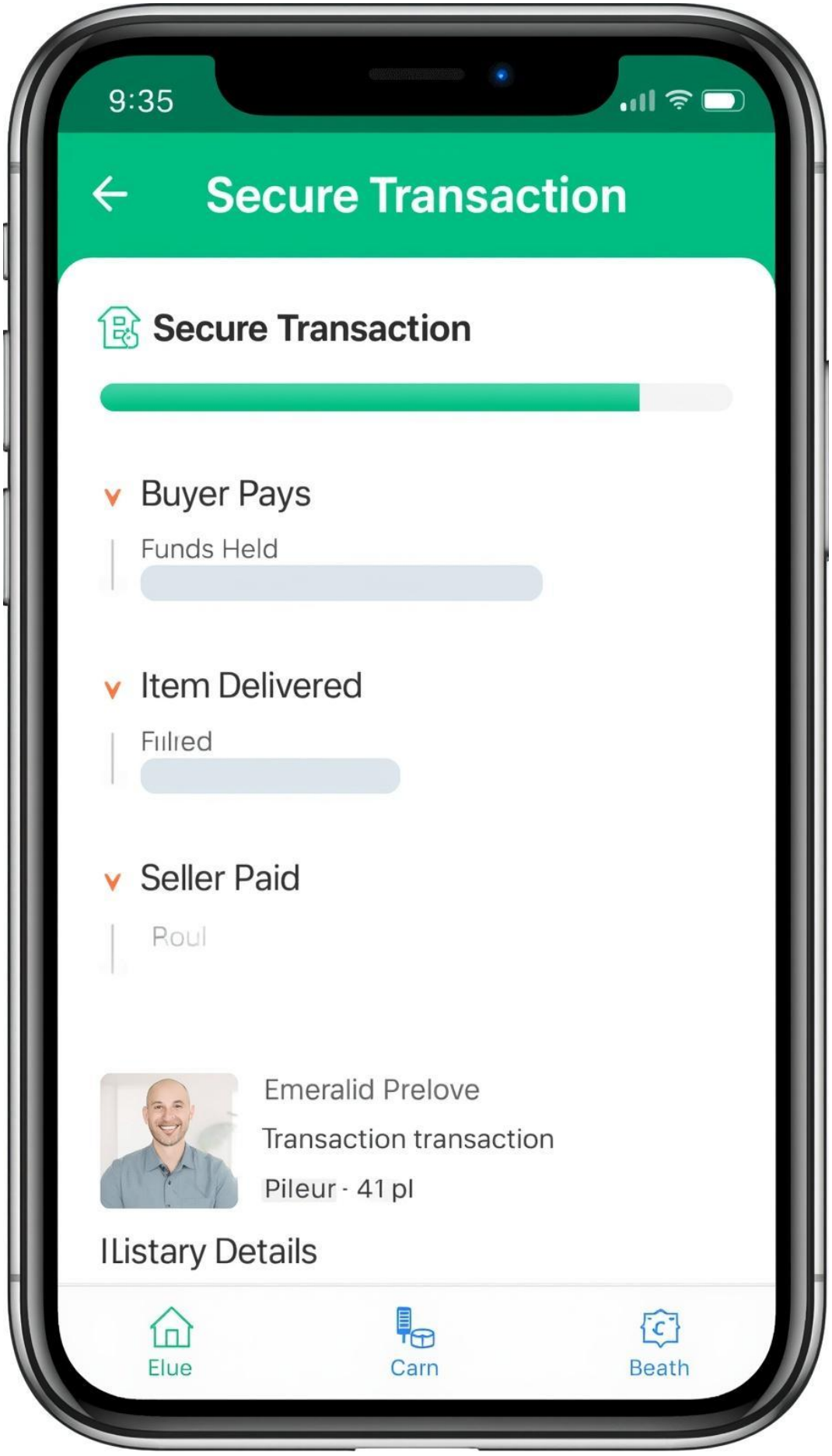


Who it benefits: By facilitating intuitive interaction without the need for specific instructions, this helps student users. By offering these visual clues, the interface makes sure that a student entering the marketplace for the first time knows right away that tapping a card will open a listing, which lowers cognitive friction and improves task efficiency (Norman, 2013; Rogers, Sharp and Preece, 2023, p. 31).

3. Design Principle 2: Mapping

Definition and Motivation (Why): The relationship between controls and their consequences on the world is referred to as mapping (Rogers, Sharp, and Preece, 2023, p. 30). Effective mapping in interaction design guarantees that an action's outcomes are predictable and follow a logical, natural sequence that corresponds with the user's preexisting mental model of how a task should proceed (Rogers, Sharp, and Preece, 2023, p. 30).

Implementation (How): A horizontal linear progress bar (B) in Figure 2 (Low-Fidelity Secure Transaction Screen) shows the escrow payment status. The order is left-to-right: The buyer pays, the funds are held, the item is delivered, and the seller is paid.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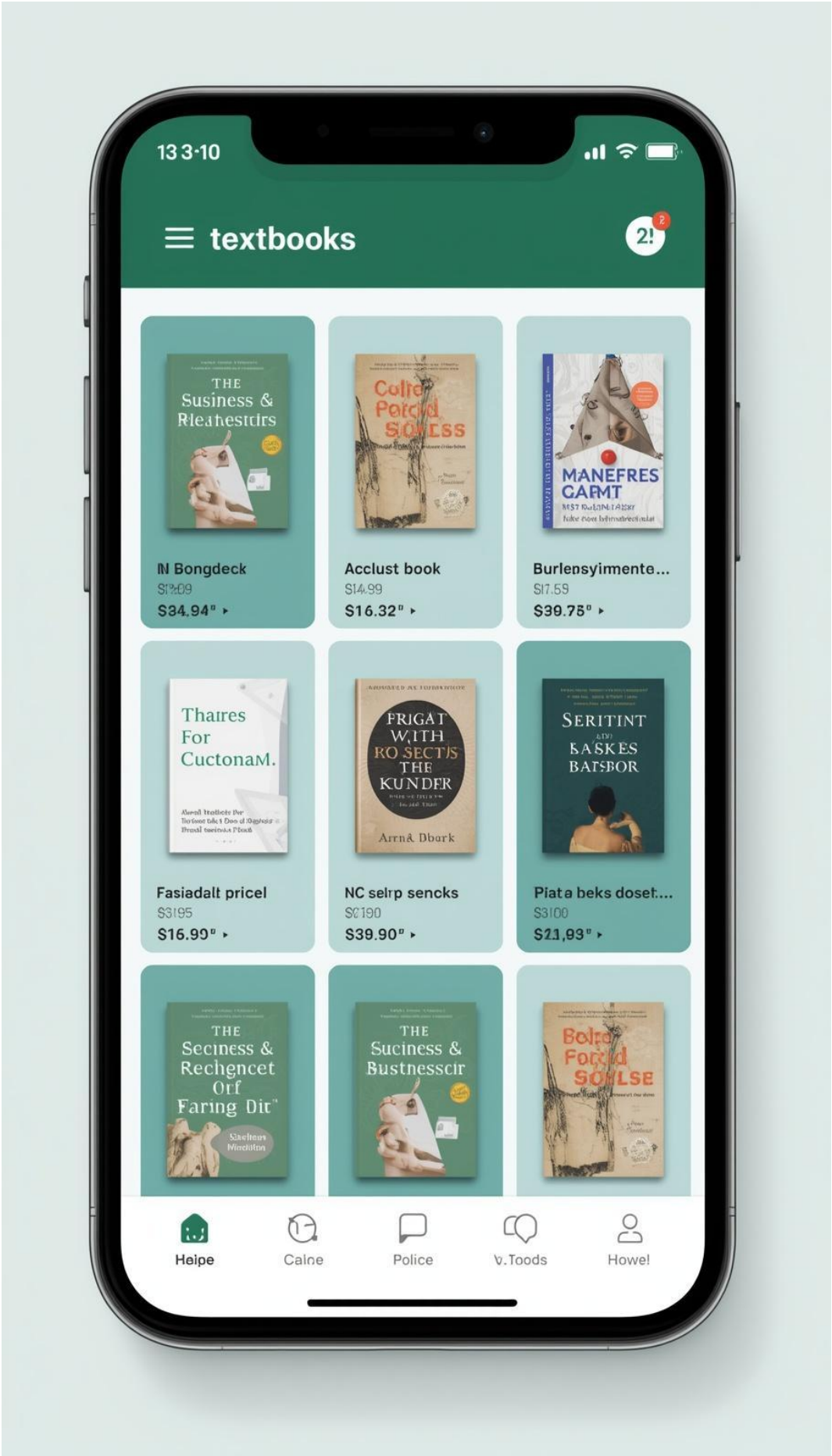
Who it benefits: This replicates the timeline of a real-world transaction, which is advantageous to both student buyers and sellers. This layout offers a natural mapping that lessens anxiety during high-stakes P2P transactions since users often think of time and progress as going from left to right. Trust in the platform is greatly enhanced by guaranteeing predictable system behavior (Norman, 2013; Strategic Integration…, 2025).

4. Design Principle 3: Simplicity

Definition and Motivation (Why): For mobile platforms, simplicity is a crucial design philosophy that entails eliminating superfluous buttons, icons, and visuals to produce a crisper, cleaner interface (Rogers, Sharp, and Preece, 2023, p. 31). Many designers make the error of trying to fit too much information onto one screen, which makes it unsuitable for users to locate what they need (Rogers, Sharp, and Preece, 2023, p. 31).

Implementation (How): The dashboard uses a Bento Grid Layout (C), as seen in Figure 1. Complex branding, supplementary metadata, and unnecessary auxiliary buttons on the main stream are deliberately left out of this modular approach. The interface only shows the most current entries and the main book categories (such Science and Business).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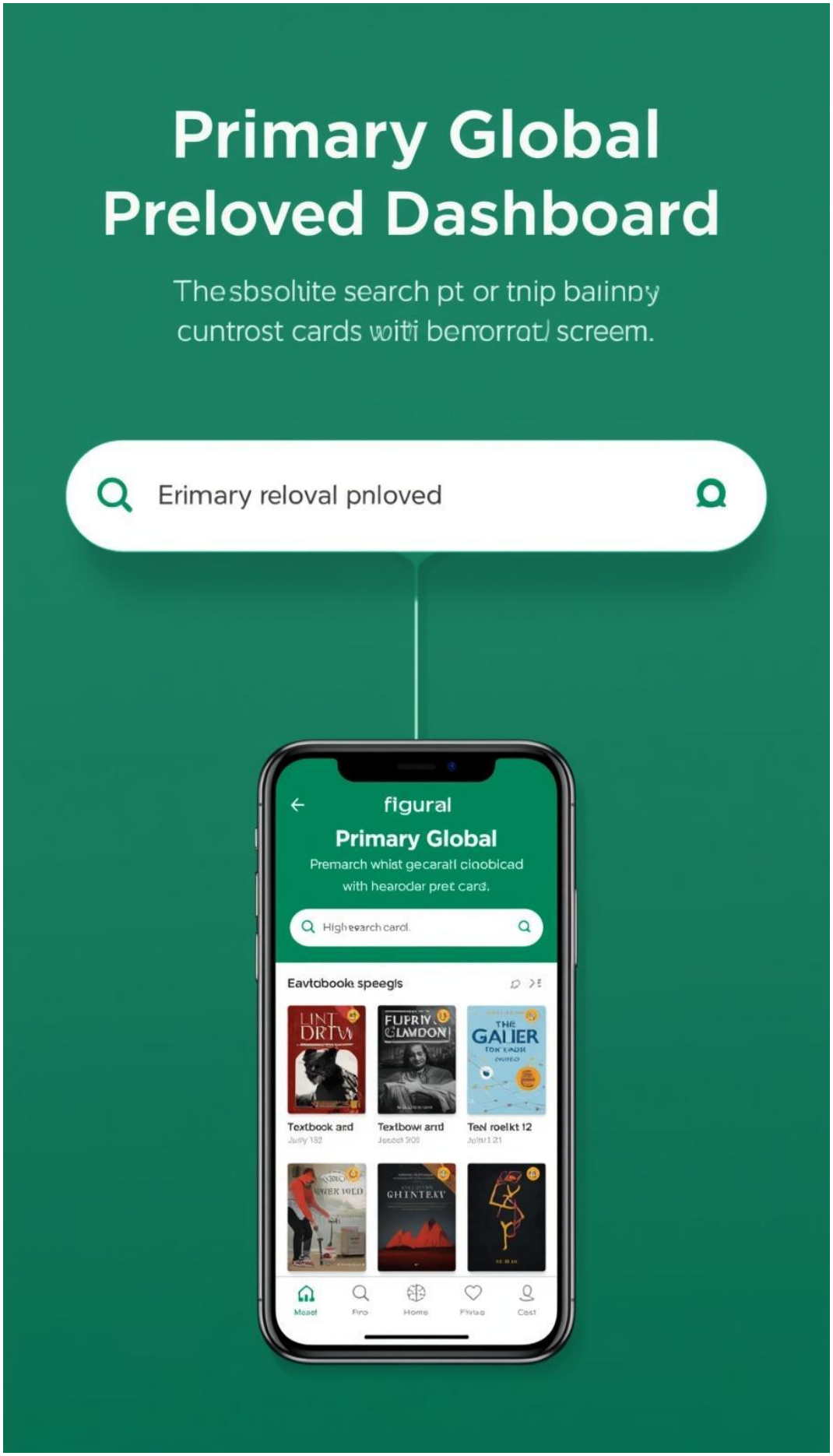
Who it benefits: The university student who is constantly on the go between courses can benefit from this. The interface is nonetheless very scannable on a small mobile device by offering just the right amount of information. In time-constrained college settings, this lessens the user's cognitive load and facilitates effective task completion (Tidwell, Brewer, and Valencia, 2020; Rogers, Sharp, and Preece, 2023, p. 31).

5. Design Principle 4: Findability

Definition and Motivation (Why): The term findability describes how simple it is to find or locate a specific item or piece of information within a system (Rogers, Sharp, and Preece, 2023, p. 27). This is particularly important for marketplaces where consumers frequently enter with a certain search intent(Rogers, Sharp, and Preece, 2023, p. 27).

Implementation (How): The main global search bar (D) in Figure 1 occupies the entire width of the screen and is located at the very top of the interface. It is the most visually noticeable component on the dashboard thanks to its high-contrast border.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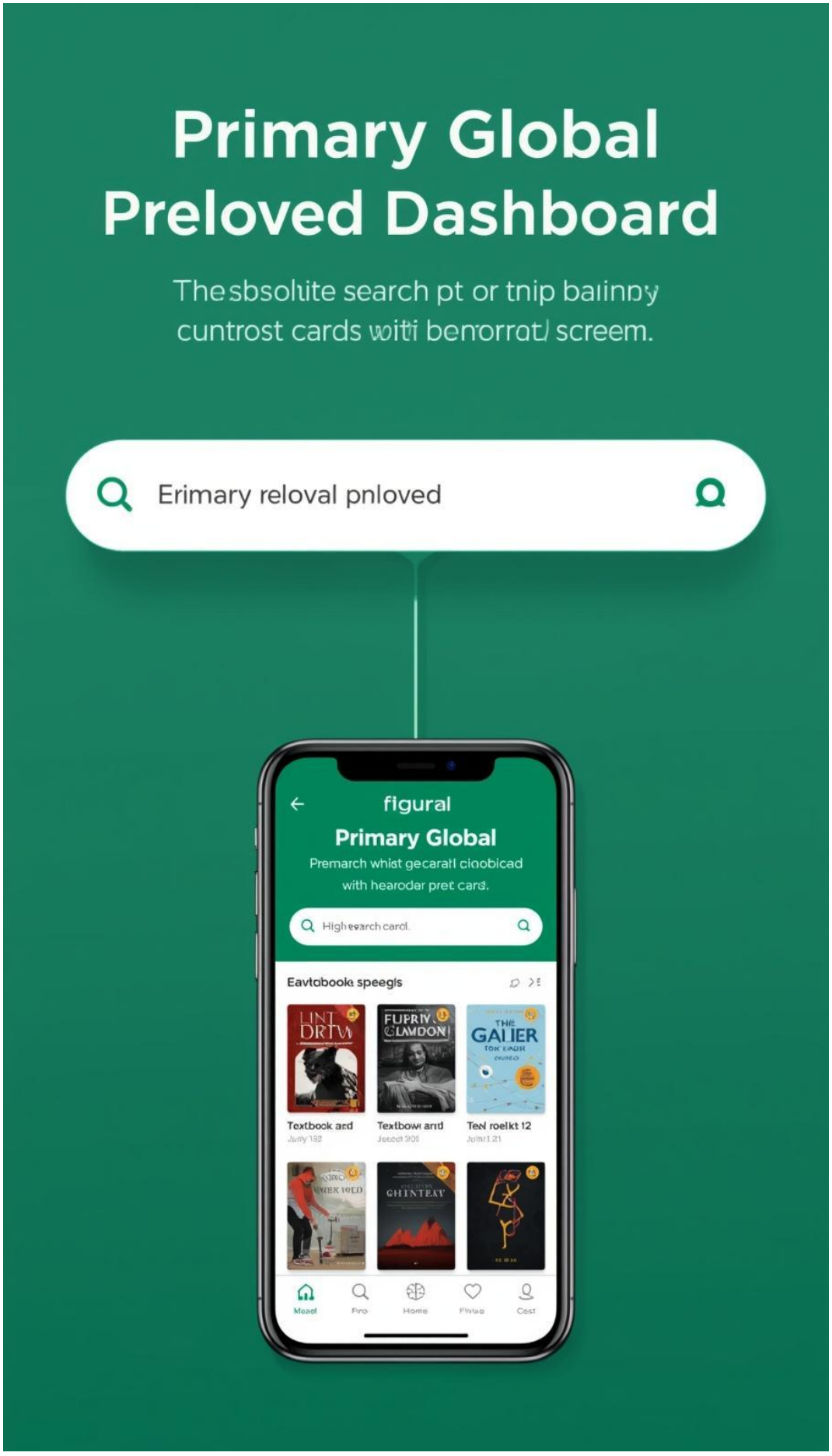
Who it benefits: Task-oriented pupils who have a certain textbook title or ISBN in mind will profit from this. Because the student does not need to hunt for the marketplace entry point, the software improves system efficiency and decreases user irritation by making sure the search functionality is unmissable (Morville, 2005; Rogers, Sharp and Preece, 2023, p. 27).

6. Design Principle 5: Navigability

Definition and Motivation (Why): In an interface, navigability refers to how clear it is what to do and where to go (Rogers, Sharp, and Preece, 2023, p. 27). It entails designing menus and navigation routes so that users may navigate the information architecture without becoming lost (Rogers, Sharp, and Preece, 2023, p. 27).

Implementation (How): The prototype uses breadcrumb navigation labels (E) at the top of the content area, Science > Biology > Genetics, as shown in Figure 3 (Low-Fidelity Category Screen).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Who it benefits: Students gain from this when navigating the complex, hierarchical structures of academic disciplines. One-click access to higher tiers of the application is made possible by breadcrumbs, which offer a persistent map of the user's current location. This guarantees a logical flow across the category system and avoids the confusion frequently associated with multi-layered mobile navigation (Tidwell, Brewer, and Valencia, 2020; Rogers, Sharp, and Preece, 2023, pp. 224, 589).

7. Evaluation Plan

Motivation (Why): In order to make sure the product is suitable and acceptable, evaluation is a crucial procedure that gathers information on how users interact with a prototype (Rogers, Sharp, and Preece, 2023, pp. 519–520). Prior to final development, the objective is to discover usability obstacles and design pain points (Rogers, Sharp, and Preece, 2023, p. 520).

Method 1: Usability Testing -Controlled Setting

Definition and Motivation: By seeing typical users (students) complete typical tasks, usability testing determines how well a design supports user performance (Rogers, Sharp, and Preece, 2023, p. 526). This offers quantitative performance metrics such as completion times and error rates (Rogers, Sharp, and Preece, 2023, p. 526).

Implementation (How):

Participants: In order to fulfill the industry standard for identifying 80% of usability concerns, a sample of six (6) students will be selected (Rogers, Sharp, and Preece, 2023, pp. 426, 551).
The assignment is to use the ISBN scanner to list your 'Calculus 1' textbook for R400 (Rogers, Sharp, and Preece, 2023, p. 526).
Data collection: Video recordings of interactions will be made, and system logs will track failures and time spent on tasks (Rogers, Sharp, and Preece, 2023, p. 526).

Ethics: An informed consent form outlining their right to withdraw and how their data will be stored must be signed by each participant (Rogers, Sharp, and Preece, 2023, pp. 277–278).

Method 2: Heuristic Evaluation (Inspection Method)

Definition and MotivationIn order to detect issues based on their understanding of usability principles, specialists role-play typical users using a technique known as "discount usability" (Rogers, Sharp, and Preece, 2023, p. 584). Early in the design cycle, it is quite successful in identifying clear navigation and consistency problems (Rogers, Sharp, and Preece, 2023, p. 584).

Implementation (How):

Participants: The prototype will be separately examined by three (3) UX specialists (Rogers, Sharp, and Preece, 2023, p. 584).
Procedure: The app will be run through twice by experts. The UI is assessed using Nielsen's 10 Heuristics, particularly Visibility of System Status and Error Prevention (Rogers, Sharp, and Preece, 2023, pp. 585, 590). The first pass gives a sense of the flow.
Result: The final redesign will be guided by a prioritized list of issues ranked by severity rating (0–4) (Nielsen, 1994; Rogers, Sharp, and Preece, 2023, p. 586).

Reference List

1. Morville, P. (2005) *Ambient Findability*. Sebastopol: O'Reilly Media.
2. Nielsen, J. (1994) 'Heuristic Evaluation', in Nielsen, J. and Mack, R. (eds.) *Usability Inspection Methods*. New York: Wiley.
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4. Rogers, Y., Sharp, H. and Preece, J. (2023) *Interaction Design: Beyond Human-Computer Interaction*. 6th edn. Hoboken: Wiley.
5. *Strategic Integration of UX Design Patterns and Trust Architectures in University Peer-to-Peer Textbook Marketplaces: A Comprehensive Analysis for the 2025-2026 Academic Cycle* (2025) .
6. Tidwell, J., Brewer, C. and Valencia, A. (2020) *Designing Interfaces*. 3rd edn. Sebastopol: O'Reilly Media.